Partial affine-plane doubling for 3-SAT

Exact certificates and two improved time–space bounds

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October 27, 2026 (UTC)

Abstract

We review the thirty-four refinement packets present in ProveIt's SubexponentialSAT project through commit `324f3d9b009a`, together with the OpenAI manuscript and the public Anthropic Lean development. The latest five packets already contain affine-plane menus, coupled support certificates, rectangular backends, and joint time/storage selection. The new contribution is a controlled partial duplication of the nine-point affine plane and a mixed-arity recursion that pays for every child entry.

Relative to the corpus's stated exact contraction interface, the new fifteen-group construction gives time and space coefficients

$$C_T = \frac{\pi}{\ln 2}\sqrt{\frac{32012591492467}{5556600000000}} = 10.87878166668\ldots, \qquad C_S = C_T\sqrt{\frac{200}{231}} = 10.12253456921\ldots$$

on the scale $L/\sqrt{\log_2 L}$ for 3-SAT, where $L \le 3m$ is the normalized literal-occurrence count. A fully doubled eighteen-group construction gives the alternative pair

$$(C_T, C_S)_{18} = (11.07682383502\ldots, 9.97733739357\ldots).$$

The first time coefficient is about 3.42% below the strongest compatible composition of prior ingredients constructed in this review, $11.26420175559\ldots$, and about 18.69% below the smallest printed comparable low-degree time coefficient.

The finite geometry and joint selectors have independent exact checks; the uniform recursion and 3-SAT translation are proved here. The new code is not a complete SAT solver or a proof-assistant formalization. The unrestricted n-only order $n \log\log n/\sqrt{\log n}$ remains unchanged. A breakthrough in logarithmic order has not been established.

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1 Results, comparisons, and proof status

The problem throughout is ordinary 3-SAT. Exact counting is used as a method for deciding satisfiability. Let n be the number of declared variables, m the number of clauses, and L the number of literal occurrences after removing tautologies and duplicate literals within a clause. Numerical exponent coefficients use base-two logarithms; ln denotes the natural logarithm.

Theorem 1.1 (Explicit occurrence bounds for 3-SAT)**.** *Under the inherited exact contraction premises in Section 3, there are deterministic algorithms deciding 3-SAT, and in fact returning its exact number of satisfying assignments, with*

$$T \le \mathrm{poly}(n,m)\, 2^{(C_T+o(1))L/\sqrt{\log_2 L}}, \qquad S \le \mathrm{poly}(n,m)\, 2^{(C_S+o(1))L/\sqrt{\log_2 L}},$$

for either of the following simultaneous pairs:

$$(C_T, C_S)_{15} = \frac{\pi}{\ln 2}\left(\sqrt{\frac{32012591492467}{5556600000000}}, \sqrt{\frac{32012591492467}{6417873000000}}\right),$$
$$(C_T, C_S)_{18} = \frac{\pi}{\ln 2}\left(\sqrt{\frac{563681831683}{94374000000}}, \sqrt{\frac{14000}{2889}}\right).$$

The error terms are uniform over normalized formulas. Empty and bounded-size cases are handled directly. The pairs are alternative algorithms.

The normalized contraction theorem starts with at most s wires, maximum degree $\lceil s^{1/12} \rceil$, and precision and truncation order at most s. Section 6 proves that fixing variables with more than $\lceil \log_2(L+1) \rceil$ occurrences meets this degree condition at lower-order cost, with $s = L + 1$. This threshold choice composes an existing incidence construction; it is not a new SAT encoding.

Comparable low-degree cached bound	Time	Space
Smallest printed low-degree time certificate [8]	13.379649	13.253025
Strongest prior-ingredient composition constructed in this review	11.264202	11.264202
New partial doubling, $15 \to 10 \to 7 \to 4$	**10.878782**	10.122535
New full doubling, $18 \to 9$	11.076824	**9.977337**

Applying newer ingredients to an old menu can improve a printed number without a new construction. We therefore reconstruct the stronger 11.264202 comparison before comparing the new geometry. It is a concrete certified bound, not a globally optimized prior frontier. Its equal time/space bound is safe; existing ingredients may permit further memory tradeoffs. A different printed low-degree pair has space $12.36130\ldots$ [1]. Both new algorithms use subexponential space; the latest low-degree polynomial-space coefficient $16.81223\ldots$ remains a different resource guarantee [15].

The main time coefficient is smaller than the constructed comparison by $0.38542008891\ldots$, or about 3.42%. For the corresponding upper-bound expressions U_{prior} and U_{15},

$$\log_2 \frac{U_{\text{prior}}}{U_{15}} = (0.38542008891\ldots + o(1))\frac{L}{\sqrt{\log_2 L}}.$$

This is a comparison of upper-bound functions, not a lower bound on an earlier algorithm or an empirical SAT speedup. Since $L \le 3m$, the main pair gives coefficients $(32.63634500004\ldots, 30.36760370765\ldots)$ on $m/\sqrt{\log_2 m}$. At fixed density $m \le Dn$, multiply the normalized constants by $3D$ for an $n/\sqrt{\log_2 n}$ bound. At arbitrary density retain the existing compressed and density-sensitive branches. The worst-case n-only order remains

$$\operatorname{poly}(n, m)\, 2^{O(n \log\log n/\sqrt{\log n})}.$$

No comparison is made with an unspecified constant in the independent Anthropic theorem.

Dependency boundary. The new finite inequalities, joint integer selectors, strict schedule arithmetic, recurrence analysis and occurrence-threshold composition are supplied here. The deformation, exact recovery, packing, mode-entropy and matrix-ring interfaces are imported mathematical premises. This is a research proof relative to those interfaces, not an independent end-to-end verification of the source preprint. The source repository itself records its packets as unverified claims [1].

2 Review of recent commits and the current comparison

The final snapshot is `324f3d9b009ab0323f33be65cd6577867c9c4338`. The review began at `7de9d09fec`, then incorporated the five packets in the next project-affecting commit. The relevant recent sequence is:

Commit prefix	Relevant change
`2b925d8800`	Seven first-round refinement packets.
`3e09608991`	OpenAI manuscript and quantitative appendices vendored.
`6d3adc4506`	Eight second-round packets.
`450252e9d8`	Five third-round packets.
`a7ae5ab55f`	Nine fourth-round packets and new printed leaders.
`7de9d09fec`	Public Anthropic formalization linked.
`324f3d9b00`	Five fifth-round packets; 2026-10-27 03:55:13 UTC.

The final index has thirty-four refinement packets and two primary manuscripts [1]. We surveyed all available report sources, with deeper inspection of the shared interface, comparison leaders, compatible compositions and smaller-order claims. This is not a fresh verification of every proof in all packets. The source ledger records 144 report-source files and 41 fifth-round Python files checked against their Git blob IDs. Four primary-manuscript source files and two Lean entry files are additionally identified by immutable paths and local checksums. Retrieval-integrity checks do not certify mathematics.

2.1 What the fifth round already contains

Packet	Generic time	Generic space	Relevant ingredient
`sat_pairwise_certificates`	29.81161	29.81161	Weighted coupled pairs; rectangular backend
`sat_affine_contraction`	30.25704	25.98907	Joint time/storage selector
`sat_affine_geometry`	30.26987	29.48754	Affine and punctured menus
`sat_affine_cauchy`	30.41335	29.46603	Cyclic thirteen-plan menu
`sat_affine_plane`	30.69134	29.46603	Affine-nine and a larger extension

These are per-wire coefficients at the generic degree promise, not coefficients on arbitrary SAT variables. Generic polynomial space remains $46.184845\ldots$; the first packet improves low-degree polynomial space to $16.81223\ldots$ [15]. All these branches remain available. A preliminary generic 30.669 construction from the initial review is superseded by the fifth round and is not claimed as an advance.

Affine-nine geometry, support averaging, norm-coupled inequalities, rectangular multiplication and simultaneous storage selection are all prior ingredients. The new objects are the explicitly ordered partial fifteen-point and full eighteen-point duplications and their completed low-degree schedules. Novelty is relative to the inspected corpus; no literature-wide priority assertion is made.

2.2 The public formalization

At Anthropic commit `1a6eee293b9e00cb432eef6018a21f523623a9af`, `eth/MainResults.lean` states the theorem using a deterministic finite-alphabet Mathlib multi-stack Turing machine, and `eth/Solution.lean` supplies its proof entry points [3]. For clause width three its quantitative order is

$$2^{O\left(\min\{m/\sqrt{\log m},\, n\log\log n/\sqrt{\log n}\}\right)}.$$

The input conventions include occurrence of every variable and no repeated clause lists; normalization supplies input-polynomial overhead. We inspected these statements, proof entry points and the paper, without rebuilding the full Lean development. Its public availability corrects older descriptions as unpublished. It does not formalize the new contractor or automatically transfer our explicit constants through a different proof.

3 Inherited exact interface and local evaluation model

The analysis imports the exact deformation and recovery framework of the OpenAI manuscript and the reviewed refinements [2, 5, 6, 7, 15, 16]. The following are the precise parts used.

At an active call, let M be the wire count, r the fixed group count, u the taper range, and a the cumulative packet exponent. Set

$$\kappa = \frac{\pi\sqrt{2}}{\ln 2}, \qquad K(M,u) = \kappa\frac{M}{\sqrt{u}}.$$

Nearly lossless padding gives child wire count $(2/r + o(1))M$ when the packing gates are small. With $Q = r^a$ and current side degrees R, C, sufficient gates are

$$QR/M = o(1), \qquad Q^2C \operatorname{polylog}(s)/M = o(1), \qquad 2^aQ^2/M = o(1).$$

Both retained and freshly created degrees must be tracked. For a retained valuation $i < u$, put $D = 2^{a-i}$. The guarded working precision is $(6 + o(1))MD/u$. The sharp entropy estimate comes from the joint Cauchy weights $a+1-i+4q-2\operatorname{popcount}(q)$ and at most $(2+o(1))M/(r^2D)$ families of each mode index for every directed pair.

For an exact allocation of mode-weight budgets, the remaining finite sum is

$$\sum_{\xi_{gh}\in\mathcal{I}_{gh}} \prod_{g=1}^{r} Z_g((\xi_e)_{e\ni g}). \tag{1}$$

Empty-list allocations are skipped; the number of allocations is polynomial for fixed r. Each entry of Z_g is a genuine recursive contraction, never a unit-cost oracle. Write

$$h_{gh} = \log_2 |\mathcal{I}_{gh}|, \qquad g_{\{i,j\}} = h_{ij} + h_{ji}.$$

For every fixed nonnegative directed vector z, the inherited inequality is

$$\sum_{g\neq h} z_{gh}h_{gh} \le (\|z\|_2/r + o(1))K.$$

In particular, a fixed undirected support v satisfies

$$v \cdot g \le (\sqrt{2}\|v\|_2/r + o(1))K. \tag{2}$$

We use this only for finitely many fixed supports and sums, so the uniform error applies. Exact list cardinalities are computed by the usual finite unbounded-knapsack recurrence on positive integer weights. The budget sizes and cardinality bit lengths are polynomial in s. The same finite dynamic program supplies rank/unrank operations and polynomial-delay enumeration of each mode list; these are inherited interface obligations.

The input coefficient ring is $(\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$ with $1 \le P_0, \ell \le s$. Ring rank, precision and coordinate bit lengths remain polynomial in s. Original entry routines may take $\operatorname{poly}(s+\mathcal{L})2^{O(s^{9/10})}$ time and polynomial space; the equality and clause entries used for 3-SAT have polynomial-time access. Every tensor keeps its immutable creation-time description and the canonical representatives fixed at its creation precision. Fresh tensors have degree at most twice their creation packet size, and their entry cost is bounded by an exponential in that size with polynomial factors.

Each singleton group cache has $2(r-1)$ directed incidence classes, so its entries require at most $2^{(\alpha_r+o(1))K}$ child evaluations, where $\alpha_r = \sqrt{2(r-1)}/r$. If β_r is the selected plan's local time coefficient, the complete recurrence is

$$T_d \le \operatorname{poly}(s)\left[2^{(\alpha_{r_d}+o(1))K_d}T_{d+1} + 2^{(\beta_{r_d}+o(1))K_d}\right]. \tag{3}$$

Local table storage and recursion stack storage are charged separately.

3.1 The prior rectangular backend

For sorted axis entropies $x \ge y \ge z \ge 0$, use

$$F(x,y,z) = \max\left\{(1+\varepsilon/2)(x+y),\quad x+(47/56+\varepsilon)y+(15/28)z\right\},\qquad \varepsilon = 10^{-5}. \tag{4}$$

The input/output-size term is retained. For unsorted axes the cost is the maximum of nine linear forms, whose integer coefficient triples after multiplication by $D_* = 1{,}400{,}000$ are used below. Rearrangement is valid because $1 \ge 47/56 + \varepsilon \ge 15/28$.

This backend is imported from the reviewed rectangular and weighted-pair interfaces [15]. The field bounds $\alpha > 3/10$ and $\omega < 19/8$ suffice [4]. Tensoring a fixed algorithm near the dual-exponent point with a square algorithm, then blocking the largest dimension, gives the two branches of (4), with the fixed slack ε covering the finite algorithm choice. The coordinate/CRT bridge is essential over the original finite coefficient rings: expand entries in polynomially many integer coordinates, compute over suitable odd prime fields, reconstruct bounded integers and reduce into the original ring. It does not invert a nonunit there. The backend uses space bounded by a polynomial factor times its largest input/output array. These facts are imported theory, not implementations supplied by the small algebra tests.

A plan partitions the groups into three ordered blocks. It recursively splits each block at the midpoint, multiplies the resulting boundary tables, and contracts the final three block tables. Each operation has the three edge classes of a tripartition (A, B, O), where O is the remaining vertex set. A time support assigns a rectangular coefficient to each crossing class and zero inside each part. A stored table has the zero–one boundary support of its vertex subset. This includes singleton caches, intermediate arrays and final block arrays.

3.2 The coupled-pair argument

For four plans, suppose their time-support families obey

$$\|v\|^2 + \|w\|^2 + 6v\cdot w \le A_{ij} \quad (v \in \mathcal{V}_i,\ w \in \mathcal{V}_j,\ i < j).$$

For any choices $v_j \in \mathcal{V}_j$, the exact identity is

$$\left\|\frac{1}{4}\sum_j v_j\right\|^2 = \frac{1}{48}\sum_{i<j}(\|v_i\|^2 + \|v_j\|^2 + 6v_i\cdot v_j) \le \frac{1}{48}\sum_{i<j} A_{ij}.$$

If $F_j(g) = D_*^{-1}\max_{v\in\mathcal{V}_j} v\cdot g$, then

$$\min_j F_j(g) \le \frac{1}{4}\sum_j F_j(g) = \frac{1}{4D_*}\max_{v_j\in\mathcal{V}_j}\left(\sum_j v_j\right)\cdot g.$$

Together with (2), this proves local coefficient

$$\beta^2 = \frac{2\sum_{i<j} A_{ij}}{48 D_*^2 r^2}.$$

The argument averages supports at the actual allocation and never interchanges a minimum and a maximum. Applying it to time families augmented by scaled boundary supports gives simultaneous time and storage bounds, provided every mixed pair is also certified. These methods are prior work [15, 16]; the finite supports below are new.

For independent reproduction, the inner product of two supports can also be computed from their 3×3 contingency table. If $n_{ab} = |A_a \cap B_b|$ and coefficients on distinct part pairs are c_{ab} and d_{uv}, the contribution for $a < b$ and $u < v$ is

$$c_{ab}d_{uv}(n_{au}n_{bv} + n_{av}n_{bu}).$$

Summing these nine terms counts every edge crossing both tripartitions once. Direct coordinate enumeration in the standalone checkers gives the same finite integer quantities without relying on any numerical optimizer.

4 Partial affine duplication: the main improvement

The final comparison is against all five rounds present at ProveIt commit `324f3d9b009a`. The fifth round already contains affine-nine plans, coupled support-pair bounds, the rectangular backend, and joint time/space selection [15, 16, 17, 18, 19]. We import those ingredients. The new construction duplicates six points of the affine plane, producing a fifteen-group root menu with a more favorable balance between local contraction and recursive child costs.

For the low-degree normalized interface, the resulting coefficients are

$$C_T = \frac{\pi}{\ln 2}\sqrt{\frac{32012591492467}{5556600000000}} = 10.87878166668154\ldots,$$
$$C_S = \frac{\pi}{\ln 2}\sqrt{\frac{32012591492467}{6417873000000}} = 10.12253456921954\ldots.$$

Thus 10.879 and 10.123 are safe eventual time and space coefficients on $s/\sqrt{\lceil \log_2 s \rceil}$. They refer to one actual algorithm, whose root plan is selected jointly for time and space. The separate eighteen-group construction gives the additional pair $(11.0768238350\ldots, 9.9773373935\ldots)$ when smaller memory is preferred. These pairs must not be combined into a stronger nonexistent guarantee.

4.1 The explicit root plans

Start with the nine points $3x+y$ of $\mathbb{F}_3^2$. Duplicate points $0, 1, 2, 3, 4, 5$, adding twins $9, 10, 11, 12, 13, 14$, respectively. These are the two parallel lines $x = 0$ and $x = 1$; the line $x = 2$ remains undoubled. In every affine parallel class, append each selected point's twin to the end of its block, in increasing original-point order. The four ordered plans are

$(0, 1, 2, 9, 10, 11)$	$(3, 4, 5, 12, 13, 14)$	$(6, 7, 8)$,
$(0, 3, 6, 9, 12)$	$(1, 4, 7, 10, 13)$	$(2, 5, 8, 11, 14)$,
$(0, 4, 8, 9, 13)$	$(1, 5, 6, 10, 14)$	$(2, 3, 7, 11, 12)$,
$(0, 5, 7, 9, 14)$	$(1, 3, 8, 10, 12)$	$(2, 4, 6, 11, 13)$.

Each block is recursively split after $\lfloor k/2 \rfloor$ entries. The two resulting boundary tables are multiplied, and the three final block tables are contracted. The first plan has block sizes $(6, 6, 3)$; the other three have sizes $(5, 5, 5)$. Every intermediate merge is included in the following certificate.

Let $D_* = 1400000$. Use the round-five rectangular cost with the nine integer coefficient triples consisting of the distinct permutations of

$$(1400007, 1400007, 0), \qquad (1400000, 1175014, 750000).$$

For each operation, assign the three coefficients to its three edge classes, and zero inside each of its three parts. This gives $9(15-2)=117$ time supports v per plan, in dimension $\binom{15}{2}=105$. The corresponding normalized local cost is $F_j(g)=D_*^{-1}\max_{v\in\mathcal{V}_j} v\cdot g$, where $g_{\{i,j\}}=\log_2(|\mathcal{I}_{ij}|\,|\mathcal{I}_{ji}|)$.

The exact coupled pair maxima

$$A_{ij}=\max_{v\in\mathcal{V}_i,w\in\mathcal{V}_j}(\|v\|_2^2+\|w\|_2^2+6v\cdot w)$$

are, in pair order $01,02,03,12,13,23$,

$$(A,A,A,B,B,B),\qquad A=580173494513132,\quad B=572279799215680.$$

The standalone standard-library checker reconstructs every support and checks all $6\cdot 117^2=82134$ pairs. By the imported coupled-pair identity,

$$\left\|\frac{1}{4}\sum_{j=0}^{3}v_j\right\|_2^2\le\frac{1}{48}\sum_{i<j}A_{ij}.$$

The pointwise inequality $\min_j F_j(g)\le\frac{1}{4}\sum_j F_j(g)$ and the inherited entropy bound therefore give root local time coefficient

$$\beta_{15}^2=\frac{2\sum_{i<j}A_{ij}}{48D_*^2\,15^2}=\frac{32012591492467}{98000000000000}.$$

This is a rigorous uniform support certificate. Unlike the doubled eighteen-point symmetric menu, no matching local minimax assertion is needed or claimed for the partially doubled menu.

4.2 Joint selection and its exact storage certificate

Each stored table corresponds to a subset appearing in one of the three binary block trees. Its boundary support b is one on edges crossing that subset and zero elsewhere. There are $2r-3=27$ distinct boundary supports per root plan. Write $\mathcal{B}_j$ for this boundary family and $B_j(g)=\max_{b\in\mathcal{B}_j} b\cdot g$. Set $\rho^2=231/200$ and augment every time-support family by its scaled boundary supports: use vectors v/D_* and ρb.

The time/time pair bound is the displayed A_{ij}/D_*^2. For two boundary supports, the unscaled coupled maxima are

$$\max(\|b\|^2+\|c\|^2+6b\cdot c)=(248,248,248,238,238,238).$$

For each pair these numbers B_{ij} obey $231D_*^2B_{ij}\le 200A_{ij}$. For a mixed time/boundary pair, define

$$\Delta=200A_{ij}-200\|v\|^2-231D_*^2\|b\|^2.$$

The independent checker verifies, for both orientations of every plan pair,

$$\Delta\ge 0,\qquad \Delta^2\ge 36\cdot 200\cdot 231D_*^2(v\cdot b)^2.$$

These are exact integer inequalities, and checking the first inequality before squaring is essential. They imply

$$\|v/D_*\|^2+\|\rho b\|^2+6(v/D_*)\cdot(\rho b)\le A_{ij}/D_*^2.$$

All 4374 boundary/boundary pairs and 37908 mixed pairs pass. The same coupled-pair proof now applies to the augmented families. Selecting the plan with smallest joint objective $\max\{F_j(g),\rho B_j(g)\}$ bounds both time and ρ times boundary storage by $(\beta_{15}+o(1))K$. Hence the storage coefficient is

$$\sigma_{15}^2=\beta_{15}^2/\rho^2=\frac{32012591492467}{113190000000000}.$$

The implemented specification avoids irrational comparisons. For each positive integer pair-list product d_e, set $\widehat{g}_e = (d_e - 1).\text{bit_length} = \lceil \log_2 d_e \rceil$. For plan j compute

$$\widehat{T}_j = \max_{v \in \mathcal{V}_j} v \cdot \widehat{g}, \qquad \widehat{B}_j = \max_{b \in \mathcal{B}_j} b \cdot \widehat{g},$$

and minimize the integer

$$\max\{\widehat{T}_j^2,\ 2263800000000\,\widehat{B}_j^2\}.$$

Since $\rho^2 D_*^2 = 2263800000000$, this selects the least rounded joint objective exactly. Every coordinate is rounded upward by less than one; the support family and all coefficients are fixed. Thus the selected true objective exceeds its exact minimum by at most $O(1)$, a constant additive exponent that is negligible on the scale K. Empty exact-weight lists are skipped before selection. The integer bit lengths are polynomial.

4.3 The complete fifteen–ten–seven–four schedule

Assume the common normalized tensor interface, with at most s wires, maximum root degree $\lceil s^{1/12} \rceil$, initial precision and truncation order at most s, and the inherited immutable creation-time entry routines. Put $\lambda = \lceil \log_2 s \rceil$, $B = \lceil 6 \log_2 \lambda \rceil$, $u_d = \lfloor \eta_d \lambda \rfloor$, and $a_d = \sum_{j \le d} (u_j + B)$. Use

d	r_d	η_d
0	15	567/5000
1	10	87/5000
2	7	407/20000
$d \ge 3$	4	$(1443/80000)(63/100)^{d-3}$.

At depth one, import the fifth-round weighted ten-group menu, whose exact local coefficient is

$$\beta_{10}^2 = \frac{15952347164548384057}{49000000000000000000}.$$

At depth two use the existing punctured affine-seven menu, with $\beta_7^2 = 891/2450$. In the tail use the existing four-group plan with $\beta_4^2 = 13/25$. These are compatible prior ingredients [15, 17], not new claims.

The cumulative leading packet exponents at successive stages are

$$\frac{567}{5000}, \quad \frac{327}{2500}, \quad \frac{3023}{20000}, \quad \frac{1999}{10000}.$$

All packing gates are strictly below exponent one, as certified by

$$450^{567} < 2^{5000}, \qquad 200^{327} < 2^{2500}, \qquad 98^{3023} < 2^{20000}, \qquad 5\frac{1999}{10000} < 1.$$

For a stage with alphabet r and cumulative cap A, these inequalities state $A(1 + 2\log_2 r) < 1$. Every original or previously created degree is at most $s^{A+o(1)}$ at that stage, because $1/12 < 567/5000$ and fresh degrees are at most twice their creation packet size. The three inherited padding gates QR/M, $Q^2 C\,\text{polylog}(s)/M$, and $2^{a_d} Q^2/M$ are therefore all controlled by the displayed strict bound. A fixed positive gap absorbs all logarithmic factors and guards.

Use relative padding slack λ^{-3} and stop at $H = \lceil s/\lambda \rceil$. For a noncircular depth argument, consider hypothetical tail depths up to $\log_{20/11} \lambda + O(1)$. The least taper diverges since $63/100 > 11/20$; this is also checked as $100/63 < 20/11$. The cumulative guard is $O((\log \lambda)^2)$. The strict gates consequently

hold uniformly over this hypothetical range, and the true tail child ratio is $(1/2 + o(1)) < 11/20$. An active path cannot reach beyond the hypothetical limit. This closes the depth bootstrap and shows that floors are negligible and padding errors multiply to $1 + o(1)$. The first three child ratios are respectively $2/15$, $1/5$, and $2/7$, up to a $1 + o(1)$ factor.

The inherited creation-time precision overhead satisfies $p = O(s\lambda^2 2^{a_d-1})$. For $i < u_d$, $D = 2^{a_d-i} \geq 2^{a_{d-1}+B+1}$; also $M \geq s/\lambda$ and $u_d = O(\lambda)$ at active calls. Thus this overhead is negligible relative to MD/u_d, with ratio $O(\lambda^4 2^{-B}) = o(1)$. All arithmetic bit lengths remain polynomial. Original entry work $2^{O(s^{9/10})}$, fresh-entry work $\exp(s^{1999/10000+o(1)})$, and direct leaf work $2^{O(s/\lambda)}$ are lower order. Creation-time tensor descriptions remain frozen. Products of polynomial overheads along $O(\log \lambda)$ levels add only $O(\lambda \log \lambda)$ to the time exponent.

4.4 Charging all recursive entries and storage

For every arity r, each group cache has $2(r-1)$ incident directed classes. The entropy inequality therefore charges its recursive entries with coefficient $\alpha_r = \sqrt{2(r-1)}/r$. Let

$$\mu_0 = 1, \quad \mu_1 = 2/15, \quad \mu_2 = 2/75, \quad \mu_3 = 4/525, \qquad K_d^* = \frac{\pi\sqrt{2}}{\ln 2}\frac{\mu_d}{\sqrt{\eta_d}} \quad (0 \leq d \leq 3).$$

The tail ratio of successive K_d is $\gamma = (1/2)/\sqrt{63/100} = 5/\sqrt{63}$. Unrolling the paid recurrence gives the maximum of

$$\begin{aligned} C_0 &= \beta_{15} K_0^*, \\ C_1 &= \alpha_{15} K_0^* + \beta_{10} K_1^*, \\ C_2 &= \alpha_{15} K_0^* + \alpha_{10} K_1^* + \beta_7 K_2^*, \\ C_\infty &= \alpha_{15} K_0^* + \alpha_{10} K_1^* + \alpha_7 K_2^* + \frac{\alpha_4 K_3^*}{1-\gamma}. \end{aligned}$$

Here $\alpha_4/(1-\gamma) > \beta_4$, so the last expression bounds every tail aggregation depth as well as the limiting full child-entry path to a leaf. Exact rational interval arithmetic gives

Route	C_0	C_1	C_2	C_∞
Coefficient	10.878781667	10.411338011	10.185959328	10.658059532

The displayed decimals are rounded summaries; the checkers enclose each coefficient in an exact rational interval of width 10^{-14}. Thus the root dominates. Substituting its exact local coefficient yields the stated C_T. The strict inequalities also absorb all uniform $o(1)$ losses in the imported interface.

For space the root uses $\sigma_{15} K_0^*$ by its joint certificate. All later stored tables are bounded by their local time costs $\beta_{10} K_1^*$, $\beta_7 K_2^*$, and $\beta_4 K_3^*$, followed by geometrically smaller tail terms. The exact checker verifies these are strictly below $\sigma_{15} K_0^*$. Live ancestor and descendant tables occupy a sum of sizes, so their exponent coefficients take a maximum, not a sum. This proves the stated $C_S = C_T\sqrt{200/231}$.

4.5 Comparison and the scope of the advance

For a stronger comparison than the printed tables, the audit recomposes only existing fifth-round ingredients: cyclic thirteen-group plans with the rectangular backend, followed by the existing ten-, seven-, six-, and four-group menus [15, 17, 18]. A strict rational schedule gives

$$C_{\text{prior}} = \frac{\pi}{\ln 2}\sqrt{\frac{511416908935241}{82798613625000}} = 11.26420175559485\ldots.$$

Its arities are $(13, 10, 7, 6, 4)$, its first four taper fractions are 2051/20000, 113/4000, 407/20000, and 217/20000, and its tail starts at 14023/1000000 with ratio 63/100. The accompanying independent checker reconstructs the recomposed cyclic thirteen-group certificate and verifies every packing gate and paid branch. The other displayed prior local bounds are explicit imported inputs to that comparison.

The new $10.878781\ldots$ coefficient is about 3.42% smaller than this stronger old-ingredient composition. This compares certified upper bounds, not observed runtimes or lower bounds on previous algorithms. The affine orbit exploration checked thirteen representative nonempty duplicated-point subsets under the 432 affine automorphisms. It found no stronger completed schedule among the nearby candidates examined; it does not establish global optimality over other menus or schedules.

The exponent order is unchanged. Through the independently proved occurrence-threshold branching bridge, the low-degree theorem applies to 3-SAT using the original literal-incidence budget $s = L+1$, with subleading branching cost. The prior density-sensitive, generic, and polynomial-space branches remain available. No smaller unrestricted n-only logarithmic order has been established in this iteration.

5 Full doubling: a lower-memory alternative

The newly placed round-5 packets change the comparison baseline. The weighted coupled-pair certificate of *sat_pairwise_certificates* and the joint time–space selector of *sat_affine_contraction* are used here as prior methods. In particular, the nine-point affine construction is no longer a new contribution. The new object is its explicitly ordered eighteen-point doubling and the compatible low-degree composition below.

Identify point $3x + y$ with $(x, y) \in \mathbb{F}_3^2$ and add its twin $9 + 3x + y$. Order each line by its three original labels, followed by their three twins, and split blocks by the same midpoint rule. The four plans are

$$\begin{array}{lll} (0,1,2,9,10,11) & (3,4,5,12,13,14) & (6,7,8,15,16,17), \\ (0,3,6,9,12,15) & (1,4,7,10,13,16) & (2,5,8,11,14,17), \\ (0,4,8,9,13,17) & (1,5,6,10,14,15) & (2,3,7,11,12,16), \\ (0,5,7,9,14,16) & (1,3,8,10,12,17) & (2,4,6,11,13,15). \end{array}$$

Let $D_* = 1{,}400{,}000$. The rectangular backend has nine integer supports per operation, the distinct permutations of

$$(1400007, 1400007, 0), \qquad (1400000, 1175014, 750000).$$

For each of the four doubled affine plans, let $\mathcal{T}_j$ be its 144 time masks and $\mathcal{B}_j$ its 33 zero–one boundary masks. The boundary family includes every singleton, two-vertex table, three-vertex table, and six-vertex block table; the last output boundary is already represented by the complementary final block.

Write

$$A = 811701837623520, \qquad \rho^2 = \frac{A}{336 D_*^2} = \frac{1691045495049}{1372000000000}.$$

The independently reconstructed supports satisfy, for every two distinct plans and all corresponding supports,

$$\|v\|^2 + \|w\|^2 + 6v \cdot w \le A, \qquad v \in \mathcal{T}_i,\ w \in \mathcal{T}_j, \tag{5}$$

$$\|b\|^2 \le 72, \qquad b \cdot c \le 32, \qquad b \in \mathcal{B}_i,\ c \in \mathcal{B}_j. \tag{6}$$

Every one of the six time-pair maxima in (5) equals A. For a direct formula for this value, put $y = (1400000, 1175014, 750000)$. A final mask has squared norm $36\sum y_i^2$, and two transverse final masks have dot product $8(\sum y_i)^2$. Thus

$$A = 72\sum_i y_i^2 + 48\left(\sum_i y_i\right)^2.$$

Mixed time–boundary supports obey the additional bound

$$\|v/D_*\|^2 + \rho^2\|b\|^2 + 6\rho(v/D_*)\cdot b \le A/D_*^2. \tag{7}$$

This is certified without approximating ρ: set

$$D = A(336 - \|b\|^2) - 336\|v\|^2.$$

For each mixed pair the verifier checks the exact integer inequalities

$$D \ge 0, \qquad D^2 \ge 36\cdot 336\cdot A\,(v\cdot b)^2.$$

After taking nonnegative square roots these are precisely (7). The finite audit checks 124,416 time pairs, 6,534 boundary pairs, and 57,024 mixed pairs, with no violation.

For illustration, the largest structured final-operation mixed case has a short algebraic explanation. Put $S = (y_1 + y_2 + y_3)/D_*$ and $a = 36\sum_i (y_i/D_*)^2$. Then $A/D_*^2 = 2a + 48S^2 = 336\rho^2$. Its mixed expression is $a + 72\rho^2 + 96S\rho$, and the explicit nonnegative difference is

$$A/D_*^2 - (a + 72\rho^2 + 96S\rho) = 24(2\rho - S)^2.$$

The exhaustive mixed checks also include all intermediate operations and smaller boundaries.

For nonnegative entropy vector g, define the actual local exponents

$$F_j(g) = D_*^{-1}\max_{v\in\mathcal{T}_j} v\cdot g, \qquad G_j(g) = \max_{b\in\mathcal{B}_j} b\cdot g, \qquad J_j(g) = \max\{F_j(g), \rho G_j(g)\}.$$

The support family for J_j is $\mathcal{T}_j/D_* \cup \rho\mathcal{B}_j$. For any selection of one vector z_j from each such family,

$$\left\|\sum_{j=1}^{4} z_j\right\|^2 = \frac{1}{3}\sum_{i<j}(\|z_i\|^2 + \|z_j\|^2 + 6z_i\cdot z_j) \le \frac{2A}{D_*^2}.$$

Here the time pairs use (5), boundary pairs use $\rho^2(2\cdot 72 + 6\cdot 32) = A/D_*^2$, and mixed pairs use (7). The support-averaging and inherited entropy estimate therefore give a simultaneous local time–space pair

$$\beta^2 = \frac{A}{4D_*^2\cdot 18^2} = \frac{563681831683}{1764000000000}, \qquad \sigma^2 = \frac{\beta^2}{\rho^2} = \frac{7}{27}. \tag{8}$$

The time certificate is exact for this fixed menu under the Euclidean entropy relaxation. Let g be the sum of one final-operation support with coefficient order $(1400000, 1175014, 750000)$ from each plan. The independent time checker verifies

$$\|g\|^2 = 1623403675247040 = 2A, \qquad \max_{v\in\mathcal{T}_j} v\cdot g = 405850918811760 = A/2$$

for each plan, including all internal operations. Thus the normalized local time bound is attained in that relaxation. No realizability by an actual SAT input, or lower bound for a different algorithm, is asserted.

An executable integer selector. Exact comparison of expressions involving logarithms or an irrational coefficient is unnecessary. For every nonempty exact mode-list product $D_e = |\mathcal{I}_{gh}|\,|\mathcal{I}_{hg}|$, compute

$$\widehat{g}_e = (D_e - 1).\texttt{bit_length}() = \lceil \log_2 D_e \rceil.$$

This equals zero for $D_e = 1$, and satisfies $g_e \le \widehat{g}_e < g_e + 1$. Define the integers

$$\widehat{T}_j = \max_{v \in \mathcal{T}_j} v \cdot \widehat{g}, \qquad \widehat{B}_j = \max_{b \in \mathcal{B}_j} b \cdot \widehat{g}.$$

Choose the first plan minimizing the integer score

$$\max\{336\widehat{T}_j^2,\ A\widehat{B}_j^2\}. \tag{9}$$

This exactly minimizes $J_j(\widehat{g})$, since its square times $336D_*^2$ is the score. Because the finite support families and $r = 18$ are fixed, $0 \le J_j(\widehat{g}) - J_j(g) \le C_0$ for one constant C_0. Thus the selected plan satisfies

$$J_{\text{selected}}(g) \le \min_j J_j(g) + C_0.$$

The additive constant is lower order in the exponent and preserves both coefficients in (8). This selector, implemented as `select_joint` in the independent checker, is the algorithm used for the joint theorem. A selector that minimizes time alone is not asserted to have the sharper space bound.

5.1 The low-degree $18 \to 9$ composition

Under the normalized root degree premise $\Delta \le \lceil s^{1/12} \rceil$, take

$$r_0 = 18, \quad u_0 = \left\lfloor \frac{107}{1000}\lambda \right\rfloor, \qquad r_d = 9, \quad u_d = \left\lfloor \frac{29}{1600}\left(\frac{3}{8}\right)^{d-1}\lambda \right\rfloor \quad (d \ge 1).$$

Use the same logarithmic guards, immutable precision conventions, near-unit padding, and leaf threshold s/λ as in the audited interface. The total leading budget is $107/1000 + (29/1600)/(1-3/8) = 17/125$. The exact capacity witnesses are

$$648^{107} < 2^{1000}, \qquad 162^{17} < 2^{125}.$$

The root budget exceeds $1/12$ and all cumulative budgets increase, so original vertices and all retained older packets are dominated by the latest packet scale after the root. These strict capacity margins cover the resulting symmetric packing gates and all additive guards. A tail coarse child ratio $1/4$ and geometric taper ratio $3/8$ give a growing least taper through the hypothetical active depth $\log_4 \lambda + O(1)$, closing the usual noncircular depth argument.

Use the new joint selector at the root and the prior round-5 nine-point joint selector in the tail. The latter has local time $1/\sqrt{3}$ and local space $\sqrt{7/27}$. This slightly weaker tail time bound is already enough, so no new tail rectangular certificate is needed for this theorem. For this schedule, redefine its normalized scales by

$$K_0^* = \frac{\kappa}{\sqrt{107/1000}}, \qquad K_1^* = \frac{\kappa}{9\sqrt{29/1600}}.$$

With $\alpha_{18} = \sqrt{34}/18$, $\alpha_9 = 4/9$,

$$\gamma = \frac{2/9}{\sqrt{3/8}} = \sqrt{\frac{32}{243}}, \qquad x = \frac{K_1^*}{K_0^*} = \sqrt{\frac{856}{11745}},$$

the tail's paid contribution is at most

$$K_0^* \left(\alpha_{18} + x \max \left\{ \frac{1}{\sqrt{3}}, \frac{\alpha_9}{1-\gamma} \right\} \right).$$

Exact rational bounds give

$$\alpha_{18} < \frac{13}{40}, \quad x < \frac{27}{100}, \quad \gamma < \frac{23}{63}, \quad \frac{1}{\sqrt{3}} < \frac{7}{10}, \quad \frac{\alpha_9}{1-\gamma} < \frac{7}{10}.$$

Consequently this branch is below $(257/500)K_0^* < (14/25)K_0^* < \beta K_0^*$. Every child entry is therefore paid, and root local work dominates.

Since $x < 1$ and $\gamma < 1$, the largest local storage exponent is also at the root. The boundary exponent of simultaneous ancestor tables adds in storage size, not in the exponent; there are only $O(\log \lambda)$ levels. This gives

$$C_T = \frac{\pi}{\ln 2} \sqrt{\frac{563681831683}{94374000000}} = 11.0768238350204\ldots,$$
$$C_S = \frac{\pi}{\ln 2} \sqrt{\frac{14000}{2889}} = 9.97733739357645\ldots.$$

These coefficients multiply $s/\sqrt{\lambda}$ in the respective base-two exponents. They are normalized-wire coefficients under the stated degree premise, not coefficients on the number of arbitrary 3-SAT variables.

Closing-source novelty check. The five packets placed at commit `324f3d9b009a` already contain the undoubled affine plane, simultaneous boundary selection, and the weighted coupled-pair norm identity. Their mathematical sections and code were screened for doubled points, blow-ups, larger affine designs, and multiplicity constructions; no eighteen-point doubling or the foregoing $18 \to 9$ joint specialization was found. This is a bounded corpus comparison, not a priority assertion about the literature.

6 A direct and explicit bridge to 3-SAT

The occurrence construction below is already present in `eth_balanced_packing/report.tex`, Section 8 ("Consequences for SAT and limits of the improvement"), in the paragraph beginning "For a concrete occurrence-based application." That source fixes variables of degree greater than $\lceil (N+1)^{9/10} \rceil$. We give the full composition for an arbitrary root-degree promise, including a logarithmic threshold [9]. This avoids comparing a new contractor only against the weaker all-right equality-tree encoding.

Proposition 6.1 (Occurrence-preserving 3-SAT bridge)**.** *Let F be a 3-CNF, let L be its number of literal occurrences after removing tautologies and duplicate literals within clauses, and put $S = L + 1$. Suppose a uniform contraction algorithm on bipartite Boolean networks with at most S wires, root precision at most S, and maximum degree at most $d(S)$ has time bound*

$$(S + \mathcal{L})^{O(1)} 2^{(C+o(1))S/\sqrt{\log_2 S}},$$

where $\mathcal{L}$ is the description length and the $o(1)$ is uniform over admissible networks. Let an integer threshold $D(S)$ satisfy

$$\max\{3, D(S)\} \le d(S), \qquad D(S)/\sqrt{\log_2 S} \longrightarrow \infty.$$

Then 3-SAT decision, and in fact exact counting, has time bound

$$\mathrm{poly}(n, m)\, 2^{(C+o(1))L/\sqrt{\log_2 L}}.$$

If the contractor uses polynomial space, so does this algorithm. If its space coefficient is C_S on the same normalized scale, the composed space coefficient remains C_S.

Proof. An empty clause and a formula with no remaining clauses are handled directly. Let $n_0 \le L$ be the number of occurring variables. All other declared variables contribute a known factor 2^{n-n_0} to the count and do not affect the decision answer. For each occurring variable v, let d_v be its original occurrence degree, and set

$$H = \{v : d_v > D(S)\}, \qquad h = |H|.$$

Because the occurrences partition by variable, $h(D(S) + 1) \le L$. Enumerate all 2^h assignments to H.

For a branch, use the original variable–clause incidence graph after removing just the ports belonging to variables in H. Each unfixed variable carries the tensor that is one precisely when all its incident bits agree. Each clause carries its satisfaction tensor after the fixed values have been substituted. Retain every unfixed original incidence, even when a fixed literal has already made a clause identically true; the latter clause tensor is simply constant one on those ports. An isolated clause is a scalar zero or one. Retaining these incidences ensures that unfixed variables in already satisfied clauses are still counted correctly.

The network is bipartite, has

$$M = L - \sum_{v \in H} d_v \le L$$

wires, clause degree at most three, and variable degree at most $D(S)$. Its equality and clause entries are computable in polynomial time. Every assignment to the $n_0 - h$ unfixed occurring variables determines exactly one incidence assignment with nonzero equality tensors. Hence the contraction is exactly the branch's number of satisfying assignments.

Use the scalar ring $\mathbb{Z}/2^{P_0}\mathbb{Z}$ with $P_0 = n_0 - h + 1 \le S$. The branch count lies in $[0, 2^{n_0-h}]$, so its least nonnegative residue determines it exactly. Sum these counts in an $(n_0 + 1)$-bit accumulator, then multiply by the known power of two for unused declared variables. The same procedure decides satisfiability by testing whether the final count is positive.

The branching exponent obeys

$$h \le \frac{L}{D(S) + 1} = o\left(\frac{S}{\sqrt{\log_2 S}}\right).$$

All graphs and entry descriptions have size polynomial in the input. Thus multiplying the contractor's time by 2^h leaves its leading coefficient C unchanged. Finally, $S/\sqrt{\log_2 S} = (1 + o(1))L/\sqrt{\log_2 L}$. Branches are processed sequentially, so only polynomial additional storage is needed. This proves both space assertions. □

Taking $D(S) = \lceil \log_2 S \rceil$ gives logarithmic root degree and branching exponent $O(L/\log L)$, which is lower order than the contraction exponent. In particular, the bridge meets every fixed positive degree-power promise $d(S) = \lceil S^\alpha \rceil$ eventually. Alternatively, any fixed $0 < \delta < \alpha$ allows $D(S) = \lfloor S^\delta \rfloor$. Any degree promise that eventually dominates $\lceil \log_2 S \rceil$ can use the logarithmic-threshold version directly; one must still verify the degrees of the fresh tensors it creates.

Because $L \le 3m$, the occurrence bound implies

$$\mathrm{poly}(n,m)2^{(3C+o(1))m/\sqrt{\log_2 m}}.$$

For fixed $D > 0$ and $m \le Dn$, it also implies

$$\mathrm{poly}_D(n,m)2^{(3DC+o(1))n/\sqrt{\log_2 n}}.$$

The latter statement follows from eventual monotonicity of $x/\sqrt{\log_2 x}$ and $L \le 3Dn$; unused declared variables create no extra wire or precision term. These are explicit upper-bound coefficients, not measured SAT speedups. At unrestricted density, the occurrence route must be combined with the existing compressed reduction; by itself it does not improve the worst-case n-only order.

7 Why the logarithmic-order breakthrough is still missing

The new construction changes the finite contraction menu and its interaction with the global recursion. It reduces the coefficient multiplying

$$K(M,u) = \frac{\pi\sqrt{2}}{\ln 2}\frac{M}{\sqrt{u}}.$$

It does not change the dependence on u. The admissible root taper remains $u = \Theta(\log s)$, so the normalized order remains $s/\sqrt{\log s}$. The occurrence bridge then retains that denominator. The compressed reduction still incurs an $O(n \log\log n)$ incidence budget at unrestricted density [2, 12].

7.1 The useful new idea and its limitation

Partial duplication deliberately trades a slightly larger local time coefficient for a better global choice of taper and child shrinkage. The fifteen-group menu has local $\beta_{15} \approx 0.57154$, whereas the full eighteen-group menu has $\beta_{18} \approx 0.56529$. The smaller local number alone does not determine the best root bound. For fifteen groups the root taper is 0.1134λ, compared with 0.107λ for eighteen groups; the recursive entries remain cheap enough after shrinking by $2/15$. The resulting root coefficient is smaller even though its local norm certificate is larger.

Uniformly cloning still more points is not a route to a smaller order within this certificate. If every affine point has multiplicity t, then $r = 9t$. Final-block support self norms and cross products both scale as t^2, while the entropy normalization divides by $r^2 = 81t^2$. Their normalized contribution therefore stays fixed. More explicitly, if $y = (1, 47/56 + \varepsilon, 15/28)$, the final-operation coupled bound already contributes

$$\frac{1}{18}\sum_i y_i^2 + \frac{1}{27}\left(\sum_i y_i\right)^2$$

to β^2, independently of t. Internal operations can only increase the maximum used by this certificate. Meanwhile the sufficient packing capacity decreases as $1/(1 + 2\log_2 r)$. Increasing a fixed cloning multiplicity therefore does not make the coefficient vanish. Letting r grow with input size would additionally require new uniform allocation, padding and enumeration bounds; the fixed-r analysis cannot be reused without those charges.

7.2 Routes checked against the latest sources

Decision-only precision. Parity, isolation or reducing the requested output precision does not remove the current reconstruction engine's internal $\Theta(MD/u)$ precision loss. A faster decision procedure would have to avoid that recovery step with a new exact summation or reconstruction identity. This obstacle already appears in the asymptotic-analysis and precision-refinement packets [11, 14].

A thinner graded space with the same crossing decomposition. The latest weighted-pair packet proves a sharper obstruction than merely counting oscillator states [15]. Under its full-crossing-rank and polynomial-piece assumptions, a crossing on an N-symbol alphabet factors through energy windows $E \le 4h$ only if

$$N^2 \le \mathrm{poly}(h) \max_{E \le 4h} \dim W_E.$$

If $\log_2 N \sim ch^\alpha$ and $\log_2 \dim W_E \sim cE^\alpha$ with the required monotonicity, then

$$2ch^\alpha \le c(4h)^\alpha + o(h^\alpha), \quad \text{hence} \quad \alpha \ge \tfrac{1}{2}.$$

Thus simply proposing $\alpha < 1/2$ while retaining all those interfaces is inconsistent. The obstruction is scoped: a different energy window, a different reconstruction identity, a restricted image, or structured summation could fall outside it. It does not prove a lower bound for all algorithms deciding 3-SAT.

Faster matrices, more groups, or more trees. These can improve constants, but complete explicit boundary tables still pay their array sizes. A larger menu cannot remove the cost of materializing the tables it actually uses. Existing source analyses already isolate this obstruction for the full-table model [7, 15]. Likewise, increasing the group count in the dense reduction improves its degree premise without by itself removing the incidence-indexing budget. The density-sensitive and high-moment proposals already exploit that possibility [12, 1].

7.3 A concrete next target beyond coefficient optimization

The most useful next theorem would concern the *actual family* of deformed group factors generated from 3-SAT equality and clause tensors. The present entropy relaxation treats their boundary data as arbitrary arrays of the permitted sizes. A strict improvement in order needs a representation or summation rule that uses additional algebraic structure. One precise research target is a family of exact boundary representations closed under the required pair contractions and recovery operations, with construction and evaluation cost

$$2^{o(M/\sqrt{u})}\ \mathrm{poly}(s)$$

at the critical root scale. A certificate must include the representation dimension, coefficient precision, update costs and recursive entry costs. A compact description whose evaluation still enumerates all boundary indices does not meet this target.

This suggests a falsifiable program: characterize the image spaces of small genuine 3-SAT group factors after deformation; test whether a proposed exact representation survives one merge and recovery step; then prove the uniform closure and cost bounds. Full matrix rank rules out some factor-through-small-space proposals immediately, but does not itself rule out fast structured summation. No such closed representation theorem has been constructed in this report. The existing conditional $2^{O(n \log\log n/\log n)}$ target still depends on its unproved structured-mode acceleration hypothesis [14].

8 Verification, reproducibility, and remaining obligations

The archive contains complete LaTeX sources, the compiled report, exact standard-library Python checkers, deterministic JSON results, source and commit ledgers, independent audit notes, and a checksum manifest. There are two independently reconstructed certificates for the main fifteen-group geometry, with different point-class implementations.

1. `audit_round5_new15.py` constructs all four root plans and their 117 supports, checks 82,134 cross-plan time pairs, and proves the strict schedule inequalities with rational interval arithmetic. It certifies all four fully paid branches and the time and space radical identities. The ten-, seven-, and four-group local bounds are explicitly imported source inputs.

2. `check_partial_affine_joint.py` independently constructs the geometry and verifies 82,134 time pairs, 4,374 boundary pairs and 37,908 mixed pairs. It implements the rounded integer joint selector and independently verifies the complete time and space schedule.

3. `round5_affine_certificate.py` and `check_rectangular_joint.py` verify the eighteen-group alternative. The latter checks 124,416 time pairs, 6,534 boundary pairs and 57,024 mixed pairs. The former also checks the nine-group rectangular menu and a matching witness for the eighteen-group *local* Euclidean certificate; it gives no 3-SAT lower bound.

4. `audit_round5_baseline.py` independently reconstructs the prior cyclic thirteen-plan certificate with the compatible rectangular backend and checks the stronger $13 \to 10 \to 7 \to 6 \to 4$ comparison. The exact source coefficients imported at the remaining arities are identified in the audit note and dependency ledger.

5. `check_tensor_regrouping.py` compares literal directed-index summation with every affine plan in 36 fixtures, checking 144 complete plan contractions and 24,192 literal index assignments over $(\mathbb{Z}/256\mathbb{Z})[V]/(V^2)$. It includes asymmetric directed dimensions, zero factors, nonzero cases and two nilpotent factors, and checks the actual integer selectors on the fifteen- and eighteen-group fixtures. These are algebra tests, not fast-matrix benchmarks.

6. `verify_occurrence_bridge.py` compares Boolean assignment counts with incidence-wire sums on 483 small formula cases and 1,293 threshold/branch comparisons. It tests unused variables, tautologies, repeated literals, empty clauses and satisfied clauses whose remaining incidences must be retained.

The coefficient enclosures use Machin's formula with alternating-series remainders for π, a positive atanh series and geometric tail for $\ln 2$, and integer-square-root radical bounds. Floating-point numbers are used only for explicitly exploratory displays. All delivered main checks use explicit exceptions and remain active under Python's optimization flag. Ordinary and optimized replay both pass.

The finite tests establish finite identities and certificates. The proof of uniform asymptotic complexity is the schedule argument, together with the imported analytic interface. No full fast SAT solver, asymptotic matrix implementation, or new Lean/Rocq build is supplied or claimed. The finite nearby-menu exploration is recorded separately as exploratory evidence; it is not an optimality proof and is not needed by either theorem.

From the extracted packet directory, run:

```
python3 verify_manifest.py
python3 run_checks.py
python3 run_checks.py --optimized
python3 build_report.py
```

The checkers use only the Python standard library. Building the report requires pdfLaTeX and the standard packages listed in the preamble; shell escape is disabled. Replay rewrites the deterministic result files. The manifest should be checked before rebuilding the PDF, whose metadata may depend on the TeX environment. The report was compiled and visually inspected before delivery.

A Survey inventory

The following compact inventory identifies the reviewed packet families. The source ledger supplies immutable paths. Numerical records are normalized-wire coefficients; different root-degree and memory premises must be kept separate [1]. An entry here records a packet's role in the comparison and is not a new independent certification of it.

Packet	Relevant contribution or limitation
`cauchy_mode_improvement`	Factorial Cauchy weights; coefficient gain.
`eth_contraction_improvements`	Sharp taper and recovery; streamed extraction.
`3sat_precision_refinement`	Precision improvement; stronger order conditional.
`sat_density_improvement`	Density-sensitive 3-SAT order.
`eth_density_refinement`	Independently obtains intermediate-density gain.
`3sat_refinement`	Scalar parity lift; recursive interface missing.
`3sat_asymptotic_analysis`	Scoped obstructions and unconstructed interfaces.
`eth_asymmetric_packet_balancing`	Near-unit asymmetric padding.
`3sat_round2_refinement`	Packing, rank and independent-manuscript audits.
`eth_balanced_packing`	Thin gaps and the occurrence-preserving SAT bridge.
`eth_high_moment`	High-moment packing and degree-sensitive consequences.
`eth_matching_refinement`	Deterministic matching-potential packing.
`eth_packet_refinement`	Pair loads and additive precision guard.
`eth_recursive_refinement`	Recursive and annular mode refinements.
`eth_joint_cauchy`	Joint Cauchy weights of asymptotic slope four.
`sat_adaptive_contraction`	Adaptive streamed and cached aggregation.
`sat_frontier_improvements`	Variable arity and degree-sensitive bounds.
`subexponential_sat_frontier`	Variable arity and exact local kernels.
`subexponential_sat_kernel_improvements`	Extraction and separation kernels.
`subsat_contraction_tree`	Full block-table contraction and matrix-ring bridge.

Packet	Relevant contribution or limitation
`sat_adaptive_bounds`	Printed polynomial-space 46.185; low-degree cached 13.380.
`sat_adaptive_tree_certificate`	Printed generic cached 30.891/30.754.
`sat_adaptive_frontier`	Adaptive quartet and joint objectives.
`sat_adaptive_matrix_partition`	Balanced separators and adaptive triangle.
`subexponential_sat_adaptive_four`	Four-block contraction; 19/8 backend available.
`subsat_rectangular`	Rectangular contraction and precision obstructions.
`eth_adaptive_exact_kernels`	Degree-sensitive bounds and exact local kernels.
`3sat_round3_frontier`	Low-degree polynomial-space 23.666.
`eth_adaptive_schedules`	Adaptive schedules and matching-potential kernel.
`sat_pairwise_certificates`	Weighted coupled pairs; generic cached 29.812; low-degree polynomial space 16.812.
`sat_affine_contraction`	Nine-point affine geometry and joint time/storage selection; generic 30.257/25.989.
`sat_affine_geometry`	Affine and punctured menus; generic 30.270/29.488.
`sat_affine_cauchy`	Cyclic thirteen-plan menu and guarded Cauchy transforms; generic 30.413/29.466.
`sat_affine_plane`	Affine-nine and a thirteen-group extension; generic 30.691/29.466.

References

[1] Vladimir Reshetnikov, *ProveIt: SubexponentialSAT*, review pin `324f3d9b009ab0323f33be65cd6577867c9c4338`. Project index and recorded trust status.

[2] OpenAI, *The Exponential Time Hypothesis is False*, October 25, 2026. Vendored unchanged in ProveIt from upstream commit `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a`. Manuscript and quantitative appendices.

[3] Anthropic, *A $2^{o(n)}$ time algorithm for 3-SAT*, paper and Lean development at `1a6eee293b9e00cb432eef6018a21f523623a9af`. Formal statements; proof entry points; pull request 31.

[4] V. Vassilevska Williams, Y. Xu, Z. Xu and R. Zhou, *New Bounds for Matrix Multiplication: from Alpha to Omega*, SODA 2024; arXiv:2307.07970v2. `https://arxiv.org/abs/2307.07970v2`. The bounds $\omega \leq 2.371552 < 19/8$ and $\alpha \geq 0.321334 > 3/10$ suffice; no current-record claim is needed.

[5] *ETH asymmetric packet balancing.* Pinned report source.

[6] *ETH joint Cauchy refinement.* Pinned report source.

[7] *Adaptive contraction trees with exact support certificates.* Pinned report source.

[8] *Adaptive bounds for subexponential SAT.* Interface, linear flags and cached low-degree theorem.

[9] *ETH balanced packing.* Pinned report source, occurrence-based application in the SAT-consequences section.

[10] *Subexponential SAT adaptive four*. Pinned report source, including the existing 19/8 backend.

[11] *3SAT asymptotic analysis*. Decision shortcuts and the encoding obstacle.

[12] *ETH adaptive exact kernels*. Pinned report source and degree-sensitive consequences.

[13] *3SAT round-two refinement*. Pinned report source, Heisenberg dimension and locality obstructions.

[14] *Sharper precision for subexponential 3SAT*. Pinned report source, structured-mode acceleration hypothesis.

[15] *SAT pairwise certificates*, round-five packet. Pinned report, exact weighted certificates, rectangular interface and barriers.

[16] *SAT affine contraction*, round-five packet. Pinned report, including simultaneous time/storage selection and guarded mixed-arity schedules.

[17] *SAT affine geometry*, round-five packet. Pinned geometric and punctured-menu certificates.

[18] *SAT affine Cauchy*, round-five packet. Pinned report and cyclic thirteen-plan construction.

[19] *SAT affine plane*, round-five packet. Pinned report and exact finite certificates.